HyperFrog's Formal Miner Is Almost Uniform

A case study in statistically nonbinding topology predicates

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Abstract

HyperFrog [1] is an experimental LWE-style KEM whose proposed novelty is a secret-generation layer based on rejection sampling over $16 \times 16 \times 16$ voxel sets. More broadly, it provides a case study of a general structured-secret design question: when does a rejection predicate materially change the ambient secret law, and when is it statistically nonbinding? HyperFrog leaves that quantitative question open for its published parameters and for the formal acceptance predicate stated in Algorithm 1 [1]. We answer it in the negative. The popcount window $[1200, 2600]$ fails with probability 6.63×10^{-68}, and a bounded-differences argument gives $\Pr[F(s) \leq 6] < 2.18 \times 10^{-7}$, hence also $\Pr[\beta_1(s) < 8 \text{ and } 1200 \leq \mathrm{wt}(s) \leq 2600] < 2.18 \times 10^{-7}$. Consequently the Algorithm 1 secret distribution has total variation distance at most 2.18×10^{-7} from the uniform distribution on $\{0,1\}^{4096}$. Thus, at the published parameters, the formal miner does not induce a materially different topology-conditioned secret law.

1 Introduction

HyperFrog [1] proposes an LWE-style KEM in which the long-term secret is interpreted as a $16 \times 16 \times 16$ voxel occupancy field. The paper is explicit that the linear-algebraic core is standard and that the novelty lies in key-space engineering [1]: the secret is supposed to be a mined high-loop voxel object rather than an undistinguished bitstring. The same paper also states that the delicate point is whether LWE remains hard when the secret is drawn from this topology-conditioned distribution [1].

This paper uses HyperFrog as a case study of a broader structured-secret design pitfall. If a proposal begins with a simple ambient prior and then applies a structural rejection predicate, that predicate must be quantitatively binding before it can create a substantively new secret distribution. At the published HyperFrog parameter set, that premise does not become substantive for the formal acceptance predicate stated in Algorithm 1. Under that predicate, the rejection step accepts almost every uniform Bernoulli(1/2) secret. The resulting conditional distribution is within total variation distance 2.18×10^{-7} of the uniform distribution on $\{0,1\}^{4096}$, so the topological layer is operationally nonbinding at the published parameters. The relevant comparison point is therefore the existing literature on binary-secret and structured-secret LWE [2, 3, 4, 5, 6].

The original HyperFrog text is not fully consistent about connectedness: the abstract speaks of "connectivity, density, and high cycle rank," whereas the introduction makes connectedness optional and Algorithm 1 does not enforce it [1]. This note therefore adopts the formal Algorithm 1 reading and analyzes exactly that acceptance predicate. HyperFrog itself presents topology as an engineering hypothesis rather than a completed security story and explicitly raises the question of whether the conditioned distribution remains close enough to uniform for LWE hardness to plausibly transfer [1].

It also notes the acceptance-probability/min-entropy trade-off that would arise if the predicate became materially restrictive [1]. Our contribution is to answer that open quantitative question for the published parameters and formal predicate written in the paper, and to make explicit the broader design lesson that follows from the answer.

Our claim is narrow and precise. We do not claim to break standard binary-secret LWE instances. We do not claim a key-recovery attack against the underlying algebraic core. We claim instead that the published HyperFrog parameters do not instantiate a meaningful topology-conditioned secret distribution, and therefore do not realize the paper's proposed topological novelty at those parameters.

Contributions. We provide:

- a quantitative proof that HyperFrog's published acceptance event is overwhelmingly likely under the ambient Bernoulli(1/2) prior, because both the popcount window and the graph cycle-rank threshold are essentially never binding at the stated parameters,
- a general structured-secret consequence: when a rejection predicate accepts with probability $1 - \varepsilon$, the mined law is exactly ε-close to the ambient prior in total variation distance, so a statistically nonbinding predicate does not create a substantively new secret distribution,
- a parameter-level consequence for HyperFrog itself: at the published parameters, the formal Algorithm 1 miner does not produce a materially different topology-conditioned secret distribution, so the scheme's central topology-conditioned hardness question does not arise substantively in that regime, and
- an empirical confirmation over 100,000 independent samples with a fully reproducible public artifact repository [9].

2 The published HyperFrog secret distribution

Let $G = \{0, \ldots, 15\}^3$ be the 4096-voxel grid, let E_{grid} be its set of 11,520 face-adjacency edges, and let $s \in \{0,1\}^{4096}$ denote an occupancy vector. For a sampled secret s we write $V(s)$ for the number of occupied voxels, $E(s)$ for the number of occupied-occupied adjacency edges, and $C(s)$ for the number of connected components in the occupied 6-neighborhood graph. For empirical diagnostics we write $B(s)$ for the number of internal face-adjacency edges in the grid whose endpoints have opposite occupancy states (the internal boundary-edge count). HyperFrog's formal invariant is the first Betti number of the occupied adjacency graph (the graph cycle rank). Despite HyperFrog's use of the term "genus", this is cycle rank, not surface genus in the topological sense [1]. Formally,

$$\beta_1(s) = E(s) - V(s) + C(s).$$

This is the graph invariant defined and analyzed in HyperFrog's formal specification [1]. In particular, it is the cycle rank of the occupied adjacency graph, not the genus of a surface extracted from the voxels.

Throughout this paper we refer to β_1 as the graph cycle rank (equivalently, the first Betti number of the occupied adjacency graph), and use "genus" only when quoting HyperFrog's terminology.

The published acceptance event is

$$A = \{s \in \{0,1\}^{4096} : 1200 \leq \mathrm{wt}(s) \leq 2600 \text{ and } \beta_1(s) \geq 8\}.$$

For clarity, the formal Algorithm 1 miner can be paraphrased in our notation as follows [1]:

1. sample $s \leftarrow \{0,1\}^{4096}$ uniformly,
2. compute the voxel count $\mathrm{wt}(s) = V(s)$,
3. compute the graph cycle rank $\beta_1(s) = E(s) - V(s) + C(s)$ of the occupied 6-neighborhood graph,
4. accept and return s iff $1200 \le \mathrm{wt}(s) \le 2600$ and $\beta_1(s) \ge 8$,
5. otherwise repeat.

Thus the miner output is exactly the conditioned law $U \mid A$. We analyze this Algorithm 1 predicate exactly as written. HyperFrog's broader prose sometimes uses stronger language about connectedness, but connectedness is not part of the formal acceptance condition above [1]. Section 5 returns to genuinely different miners.

3 The predicate is essentially always satisfied

3.1 The popcount window is nonbinding

Proposition 1. *If s is uniform on $\{0,1\}^{4096}$, then $\mathrm{wt}(s)$ is $\mathrm{Binomial}(4096, 1/2)$ and*

$$\begin{aligned}\Pr[\mathrm{wt}(s) < 1200 \text{ or } \mathrm{wt}(s) > 2600] &= \Pr[\mathrm{wt}(s) < 1200] + \Pr[\mathrm{wt}(s) > 2600] \\ &\approx 6.4094 \times 10^{-160} + 6.6274797397 \times 10^{-68} \\ &\approx 6.6274797397 \times 10^{-68}.\end{aligned}$$

Numerically evaluating the two exact binomial tails gives the displayed value. In the public artifact code this is implemented with SciPy's binomial CDF and survival function calls. The combined probability is completely dominated by the upper tail at the stated precision. Since the mean is 2048 and the standard deviation is 32, the published interval $[1200, 2600]$ is roughly -26.5 standard deviations on the low side and $+17.25$ on the high side. Therefore the density filter contributes essentially no rejection.

3.2 The cycle-rank threshold is also nonbinding

Theorem 1. *Define $F(s) = E(s) - V(s)$. For uniform $s \in \{0,1\}^{4096}$, one has $\mathbb{E}[F(s)] = 832$. Moreover,*

$$\Pr[\beta_1(s) < 8 \text{ and } 1200 \le \mathrm{wt}(s) \le 2600] < 2.18 \times 10^{-7}.$$

Proof. Since $s \in \{0,1\}^{4096}$ is a binary occupancy vector, $V(s) = \mathrm{wt}(s)$. Each grid edge contributes 1 to $E(s)$ with probability $1/4$, so $\mathbb{E}[E(s)] = 11520/4 = 2880$. Also $\mathbb{E}[V(s)] = 4096/2 = 2048$, hence $\mathbb{E}[F(s)] = 832$. On the event $\{\mathrm{wt}(s) \ge 1200\}$, the occupied subgraph is nonempty and therefore $C(s) \ge 1$. Because $\beta_1(s) = F(s) + C(s)$, the event $\beta_1(s) < 8$ inside the popcount window implies $F(s) = \beta_1(s) - C(s) \le 7 - 1 = 6$. The 4096 coordinates of s are independent Bernoulli(1/2) bits, so McDiarmid's inequality applies to F as a function of independent variables. In the one-sided form we use, if changing coordinate i changes f by at most c_i, then for every $t > 0$,

$$\Pr[f(X) - \mathbb{E}f(X) \le -t] \le \exp\left(\frac{-2t^2}{\sum_i c_i^2}\right).$$

It therefore remains to bound the single-coordinate changes for F.

Consider flipping a single voxel x. If $d(x)$ is the degree of x in the 16^3 grid and $n_x(s) \in \{0, \dots, d(x)\}$ is the number of occupied neighbors of x, then a flip $0 \to 1$ changes V by $+1$ and E by $+n_x(s)$, so

$$\Delta F = n_x(s) - 1 \in \{-1, 0, \dots, d(x) - 1\}.$$

A flip $1 \to 0$ changes V by -1 and E by $-n_x(s)$, so

$$\Delta F = 1 - n_x(s) \in \{1 - d(x), \dots, 0, 1\}.$$

Thus in either direction

$$\Delta_x F \in [1 - d(x),\, d(x) - 1],$$

hence $|\Delta_x F| \le d(x) - 1$. The grid has 8 vertices of degree 3, 168 of degree 4, 1176 of degree 5, and 2744 of degree 6, giving

$$\sum_x (d(x) - 1)^2 = 8 \cdot 2^2 + 168 \cdot 3^2 + 1176 \cdot 4^2 + 2744 \cdot 5^2 = 88960.$$

McDiarmid's bounded-differences inequality [7] therefore yields

$$\Pr[F(s) \le 6] \le \exp\left(\frac{-2(832 - 6)^2}{88960}\right) < 2.18 \times 10^{-7},$$

which proves the claim. □

Remark on conservativeness. The following calculation is not needed for the theorem. It only calibrates how loose the McDiarmid bound is at the published parameters. The concentration bound is intentionally conservative in two distinct ways. First, the implication $\beta_1(s) < 8 \Rightarrow F(s) \le 6$ uses only the trivial lower bound $C(s) \ge 1$ on the number of connected components. Empirically, typical samples have about 58 connected components, so this reduction discards a great deal of component information before concentration is even applied. Second, McDiarmid's inequality is then used with worst-case Lipschitz constants for every voxel flip. For completeness, write $V = \sum_x X_x$ and $E = \sum_{\{u,v\} \in E_{\text{grid}}} X_u X_v$ for independent Bernoulli(1/2) bits X_x. Then $\mathrm{Var}(V) = 4096/4 = 1024$. For an edge indicator $I_{uv} = X_u X_v$, one has $\mathrm{Var}(I_{uv}) = 3/16$, disjoint edge indicators are independent, while two edges sharing one vertex have covariance $1/16$. The number of unordered edge pairs sharing a vertex is

$$\sum_x \binom{d(x)}{2} = 8\binom{3}{2} + 168\binom{4}{2} + 1176\binom{5}{2} + 2744\binom{6}{2} = 53952,$$

so

$$\mathrm{Var}(E) = 11520 \cdot \frac{3}{16} + 2 \cdot 53952 \cdot \frac{1}{16} = 8904.$$

Also

$$\mathrm{Cov}(E, V) = \sum_{\{u,v\} \in E_{\text{grid}}} \mathrm{Cov}(X_u X_v, X_u + X_v),$$

because only the two endpoints of an edge contribute nonzero covariance. For a single edge $\{u, v\}$,

$$\mathrm{Cov}(X_u X_v, X_u + X_v) = 2\,\mathrm{Cov}(X_u X_v, X_u) = 2\big(\mathbb{E}[X_u^2 X_v] - \mathbb{E}[X_u X_v]\mathbb{E}[X_u]\big) = 2\left(\frac{1}{4} - \frac{1}{8}\right) = \frac{1}{4}.$$

Hence $\mathrm{Cov}(E, V) = 11520 \cdot \frac{1}{4} = 2880$, and

$$\mathrm{Var}(F) = \mathrm{Var}(E) + \mathrm{Var}(V) - 2\,\mathrm{Cov}(E, V) = 8904 + 1024 - 5760 = 4168.$$

Thus $\mathrm{sd}(F) \approx 64.6$, and the threshold $F \leq 6$ sits about 12.8 standard deviations below the mean value 832. This confirms that the McDiarmid bound is conservative. A sharper concentration result for this degree-2 polynomial would only strengthen the conclusion, but is unnecessary for our parameter-level claim. Empirically this bound does substantially overestimate the true failure rate: in our 100,000-sample experiment we observed no violations of the published threshold $\beta_1 \geq 8$, and even the crude lower bound $E(s) - V(s) + 1$ never fell below 562.

Percolation intuition. For site percolation on the infinite simple-cubic lattice, the critical occupation probability is approximately $p_c = 0.3116077$, so Bernoulli(1/2) sampling lies well inside the supercritical regime [8]. On a finite 16^3 box this does not itself prove any particular β_1 bound, but it does explain why one should expect macroscopic occupied clusters with many cycles rather than rare high-loop outliers.

3.3 Consequences for the mined distribution

Corollary 1. *The Algorithm 1 acceptance probability satisfies* $\Pr[A] > 0.999999782$. *Moreover,*

$$A^c = \{\mathrm{wt}(s) < 1200 \text{ or } \mathrm{wt}(s) > 2600\} \,\dot\cup\, \{\beta_1(s) < 8 \text{ and } 1200 \leq \mathrm{wt}(s) \leq 2600\},$$

so

$$\begin{aligned}\Pr[A^c] &= \Pr[\mathrm{wt}(s) < 1200 \text{ or } \mathrm{wt}(s) > 2600] \\ &\quad + \Pr[\beta_1(s) < 8 \text{ and } 1200 \leq \mathrm{wt}(s) \leq 2600] \\ &< 6.63 \times 10^{-68} + 2.18 \times 10^{-7}.\end{aligned}$$

Hence the expected number of TopoMine iterations is at most 1.000000218.

Corollary 2. *Let U be the uniform distribution on $\{0,1\}^{4096}$ and let $T = U \mid A$ be the Algorithm 1 TopoMine output. Because T is obtained by conditioning a uniform distribution on the event A, the total variation distance satisfies $\Delta(U, T) = 1 - \Pr[A] = \Pr[A^c] < 2.18 \times 10^{-7}$. To see the first equality, note that for $s \in A$ one has $T(s) = U(s)/\Pr[A] > U(s)$, and for $s \notin A$ one has $T(s) = 0$, so the positive part of $(U - T)$ sums to $\Pr[A^c]$.*

Corollary 3. *Let P_{unif} be the public-key distribution obtained from binary-secret LWE with uniform $s \in \{0,1\}^{4096}$ and the same matrix/error generation as HyperFrog, and let P_{HF} be the public-key distribution obtained with $s \leftarrow T$. By data processing,*

$$\Delta(P_{\mathrm{unif}}, P_{\mathrm{HF}}) \leq \Delta(U, T) < 2.18 \times 10^{-7}.$$

These corollaries are the main quantitative result. Under the published parameters, HyperFrog does not realize a qualitatively new secret distribution: it realizes an ordinary binary secret with a conditioning step whose total variation effect is given above. The underlying mechanism is completely general: any rejection sampler that starts from a uniform prior and accepts with probability $1 - \varepsilon$ outputs a law at total variation distance exactly ε from that prior. Here the published acceptance region excludes at most about one in 4.6 million uniformly sampled secrets.

Table 1: Diagnostics for the formal Algorithm 1 predicate. Rows 1–4 are theorem-derived bounds, rows 5–11 are empirical observations from simulation.

Diagnostic	Value
Numerically evaluated failure probability of the popcount window	6.63×10^{-68}
Conservative upper bound on failure of $\beta_1 \geq 8$ inside the popcount window	$< 2.18 \times 10^{-7}$
Lower bound on TopoMine acceptance probability	> 0.999999782
Upper bound on total variation distance to uniform binary secrets	$< 2.18 \times 10^{-7}$
Certified failures of $\beta_1 \geq 8$ on 100,000 draws via $E(s) - V(s) + 1$	0/100,000
Mean cycle rank β_1 on 10,000 samples	890.11
Mean number of connected components on 10,000 samples	57.75
Connected samples ($C(s) = 1$) on 10,000 draws	0/10,000
95% upper bound for $\Pr[C(s) = 1]$	$< 3.0 \times 10^{-4}$
Mean boundary-edge count on 10,000 samples	5759.49, random expectation 5760
Minimum of $E(s) - V(s) + 1$ over 100,000 samples	562

4 Empirical confirmation

To confirm the analysis, we implemented the Algorithm 1 predicate directly in Python/NumPy/SciPy using the published 16^3 grid, uniform Bernoulli bits, 6-neighborhood adjacency, and $\beta_1(s) = E(s) - V(s) + C(s)$. We generated 100,000 independent samples for bulk statistics based on $V(s)$, $E(s)$, $B(s)$, and the lower bound $E(s) - V(s) + 1$, and 10,000 independent samples for full component analysis with exact $C(s)$ and $\beta_1(s)$. The public artifact repository [9] includes the standalone script, raw CSV outputs, JSON summaries, and the figure reproduced below.

Remark on reproducibility. Our reproduction implements Algorithm 1 directly: independent uniform bit draws, the stated popcount window, and the stated graph cycle-rank threshold. The 100,000-draw statement for $\beta_1(s) \geq 8$ is certified via the lower bound $E(s) - V(s) + 1 \leq \beta_1(s)$, while exact component counts are reported on the 10,000-draw subset. The reproduction therefore depends only on the formal paper specification and not on any repository-specific code path.

The empirical picture is consistent with the theory and illustrates the typical regime. Typical draws, which the published predicate accepts almost always, look like random half-density percolation fields: they have many connected components, very large cycle rank, and an internal boundary density equal to the random expectation. These diagnostics are consistent with ordinary half-density percolation rather than a rare strongly constrained topological object. The simulation is therefore best read as a consistency check plus descriptive evidence about typical samples, not as an independent estimate of the theorem's 10^{-7}-level failure probability. For reference, zero certified failures in 100,000 trials imply a one-sided Clopper–Pearson 95% upper confidence bound of about 3.0×10^{-5} on the empirical failure rate: this is much weaker than the theorem, but entirely consistent with it.

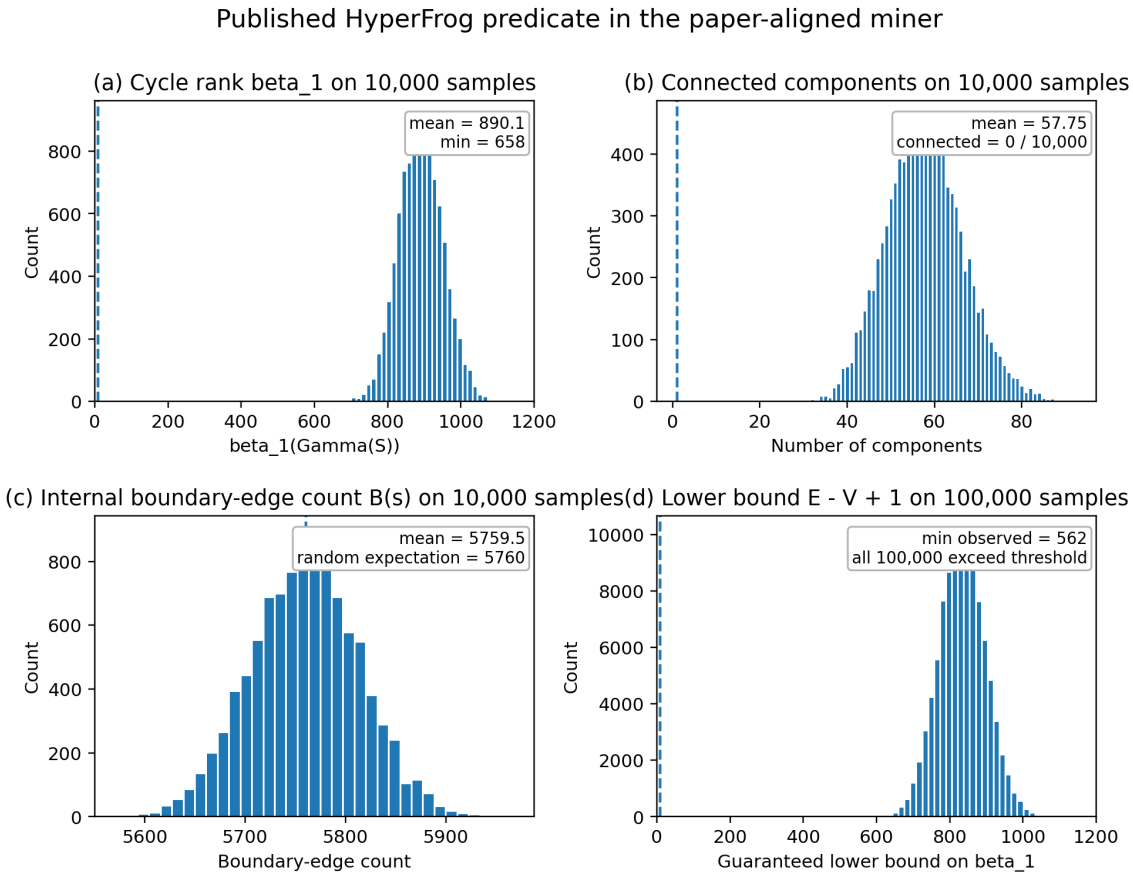

Figure 1: Diagnostics for the formal Algorithm 1 predicate. Panel (a) plots the empirical cycle-rank distribution, panel (b) the number of connected components, panel (c) the internal boundary-edge count $B(s)$, and panel (d) the lower bound $E - V + 1$. The dashed lines mark, respectively, the threshold $\beta_1 \geq 8$ in panel (a), the connectedness reference $C = 1$ in panel (b), the random-expectation baseline $B(s) = 5760$ in panel (c), and the threshold 8 for the lower bound $E - V + 1$ in panel (d). Panels (c) and (d) show that the internal boundary count matches the random baseline and that even the trivial lower bound stays far above the published threshold on all 100,000 samples.

5 Cryptographic Consequences and Design Lesson

5.1 A general lesson about rejection-sampled structure

The mechanism isolated above is not specific to topology. Any structured-secret proposal that samples from a simple ambient prior and then rejects until a predicate holds must first show that the predicate is quantitatively binding. If acceptance occurs with probability $1 - \varepsilon$, then the resulting secret law is exactly ε-close in total variation distance to the ambient prior. In that regime, the intended structure is statistically too weak to define a substantively new secret distribution. The relevant cryptographic baseline remains the ambient prior, together with whatever structured-secret analyses already exist for that baseline family.

5.2 HyperFrog at the published parameters

HyperFrog's paper states that the novelty lies in the secret-generation layer, not in the underlying LWE algebra [1]. Our theorem shows that this layer has only a very small quantitative effect in the published parameter regime. In effect, Algorithm 1 almost always returns the first uniformly sampled 4096-bit string. This sharpens, rather than contradicts, the original paper's own caveat that topology is an engineering hypothesis and that rejection sampling changes the secret distribution [1].

Accordingly, at the published parameters, the security-relevant distributional comparison is very close to that of the same public-key generation channel with unconditioned Bernoulli(1/2) secrets.

5.3 What this result does and does not show

This result is not a break of the binary-secret LWE problem. Binary-secret LWE is known to inherit decisional hardness from standard LWE through the classical reductions of Brakerski, Langlois, Peikert, Regev, and Stehlé [3] and subsequent refinements such as Micciancio's 2018 result [4]. At the same time, concrete security for small or binary secrets can deviate from estimator expectations and should be studied carefully in any concrete proposal [6]. Our contribution here is the distributional point established above: at the published parameter set, the formal Algorithm 1 miner stays statistically very close to the Bernoulli(1/2) secret law on 4096 bits.

5.4 Implications for alternative miners

One could define alternative miners either by adding connectedness as an extra rejection condition or by applying deterministic normalization after sampling. Both changes define a different scheme, not the formal Algorithm 1 miner analyzed in this paper. These two modifications behave differently distributionally. Adding connectedness as an extra rejection condition defines a new acceptance event A' and still yields the conditioned law $U \mid A'$, so the same total-variation framework applies in principle once the new acceptance probability is analyzed. By contrast, keep-largest-component is a many-to-one normalization of a uniformly sampled occupancy field, so its output law is no longer simply $U \mid A'$, and the preceding conditioning argument does not transfer directly.

More generally, one can make a topology predicate binding only by tightening the admissible set, for example by shrinking the popcount window, raising the cycle-rank threshold, or changing the grid size so that acceptance is no longer overwhelming. But that creates an explicit acceptance/entropy trade-off. If the acceptance probability is α, then rejection sampling needs $1/\alpha$ draws on average per accepted secret, while the conditioned distribution is supported on only $\alpha \cdot 2^{4096}$ strings, corresponding to a min-entropy loss of $-\log_2(\alpha)$ bits relative to the uniform 4096-bit baseline (since the conditional distribution remains uniform on A, the min-entropy is exactly $\log_2(|A|) = 4096 + \log_2(\alpha)$). A genuinely restrictive predicate therefore forces either a much lower acceptance rate, a materially smaller effective secret space, or both. At the present 16^3 Bernoulli(1/2) parameters, the natural cycle-rank scale is already in the hundreds: $\mathbb{E}[F(s)] = 832$ and the empirical mean of $\beta_1(s)$ is about 890. So a threshold of 8 sits far below the ambient regime. A materially binding predicate at this grid size would need thresholds on the scale of the natural fluctuations around those values, not single digits. Alternatively one would need a substantially smaller grid or a much narrower occupancy prior. Whether some alternative parameter regime yields a useful operating point is an open design question. The published parameters do not establish one.

Any redesigned miner would require a fresh analysis of acceptance probability, entropy loss, statistical distinguishers, and structured-secret cryptanalysis. This is exactly the regime in which LWE-with-hints and other side-information frameworks become relevant [5]. As brief descriptive context only, our keep-largest-component simulation on 10,000 samples produced mean vertex count

approximately 1965 (about 48% of the grid), mean cycle rank 889.1, and mean boundary count 5381.0. These descriptive values already indicate a nontrivial distributional shift: keep-largest-component enforces connectedness and lowers the mean Hamming weight from about 2048 under Bernoulli(1/2) to about 1965, so any security discussion would have to engage structured-secret effects directly. The near equality between the mean cycle ranks 889.1 and 890.1 also suggests that, in this regime, the largest occupied component already carries essentially all of the cycles while the discarded material contributes mostly low-cycle debris. We do not treat those numbers as a substitute for a full distributional analysis of that different miner.

6 Conclusion

The formal HyperFrog acceptance predicate in Algorithm 1 does not produce a materially different topology-conditioned secret distribution. Its popcount window is essentially never active, its cycle-rank threshold is essentially never active, and the resulting secret distribution lies within total variation distance 2.18×10^{-7} of the uniform binary distribution on 4096 bits. This is a fixed-parameter quantitative statement about the published parameter set and formal acceptance predicate, not an asymptotic negligible-function claim and not a statement about every alternative topology-constrained variant.

Accordingly, the resulting public-key distribution is extremely close to that of the same channel with unconditioned Bernoulli(1/2) secrets. HyperFrog may still be an experimental cryptosystem, but the published parameters do not realize a substantively different topology-conditioned secret space. Any future attempt to revive the idea must first define a quantitatively restrictive secret predicate, then analyze the resulting distribution against dedicated structured-secret attacks, and only then argue for novelty.

Supplementary material. The supplementary material for this paper is hosted in the public GitHub repository [9]. It contains the standalone Python reproduction, raw CSV outputs, JSON summaries, and the figure used in Section 4.

References

[1] Victor Duarte Melo. The HyperFrog Cryptosystem: High-Genus Voxel Topology as a Trapdoor for Post-Quantum KEMs. IACR ePrint 2026/195, 2026.

[2] Oded Regev. On lattices, learning with errors, random linear codes, and cryptography. *Journal of the ACM*, 56(6):34:1–34:40, 2009. https://doi.org/10.1145/1568318.1568324.

[3] Zvika Brakerski, Adeline Langlois, Chris Peikert, Oded Regev, and Damien Stehlé. Classical hardness of learning with errors. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC 2013)*, pages 575–584, 2013.

[4] Daniele Micciancio. On the Hardness of Learning With Errors with Binary Secrets. *Theory of Computing*, 14(13):1–17, 2018.

[5] Dana Dachman-Soled, Leo Ducas, Huijing Gong, and Melissa Rossi. LWE with Side Information: Attacks and Concrete Security Estimation. In *Advances in Cryptology – CRYPTO 2020*, LNCS 12171, pages 329–358. Springer, 2020.

[6] Hao Chen, Lynn Chua, Kristin Lauter, and Yongsoo Song. On the Concrete Security of LWE with Small Secret. IACR ePrint 2020/539, 2020.

[7] Colin McDiarmid. On the method of bounded differences. In *Surveys in Combinatorics, 1989*, volume 141 of *London Mathematical Society Lecture Note Series*, pages 148–188. Cambridge University Press, 1989.

[8] Junfeng Wang, Zongzheng Zhou, Wei Zhang, Timothy M. Garoni, and Youjin Deng. Bond and Site Percolation in Three Dimensions. *Physical Review E*, 87(5):052107, 2013. https://doi.org/10.1103/PhysRevE.87.052107.

[9] Mounir IDRASSI. Artifacts for *HyperFrog's Formal Miner Is Almost Uniform*. GitHub repository, snapshot at commit `6f26e95da465`. Available at https://github.com/idrassi/hyperfrog-refutation-artifacts.